

Erdős Problem #158

EP-158

L1 - TRACTABLE

OPEN

Combinatorics

ORIGINAL PROBLEM STATEMENT

Let $A \subset \mathbb{N}$ be an infinite set such that, for any n , there are most 2 solutions to $a + b = n$ with $a \leq b$.
Must $\liminf_{N \rightarrow \infty} \frac{|A \cap \{1, \dots, N\}|}{N^{1/2}} = 0$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>